

Fig. 7: Yagi-Uda antenna

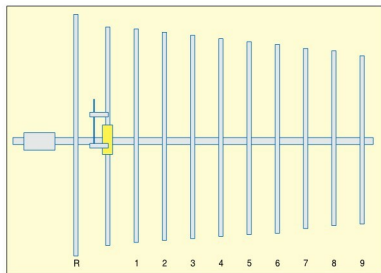


Fig. 8: Eleven-element Yagi-Uda antenna

Yagi-Uda Antenna

Frequency	Transverse dimension ($\lambda/2$)	Length for 3 elements	Length for 5 elements	Length for 15 elements
30 MHz	5 metres	6 metres	13 metres	47 metres
100 MHz	1.5 metres	1.8 metres	3.9 metres	14 metres
300 MHz	50 cm	60 cm	1.3 metres	4.7 metres
1 GHz	15 cm	18 cm	39 cm	1.4 metres
3 GHz	5 cm	6 cm	13 cm	47 cm

Yagi-Uda antenna

Yagi-Uda antenna (refer Fig. 7) is the most common terrestrial TV antenna used on the rooftops. It is usually used at frequencies between 30 MHz and 3 GHz, or a wavelength range of 10 metres to 10 cm.

The design of the antenna depends on the number of elements used in the antenna. The lengths of rods in a Yagi-Uda are about a half wavelength each, and the spacing of the elements about one-third of a wavelength.

As shown in the table, you can estimate the approximate operating

frequency of a Yagi-Uda antenna by looking at its length from afar. A Yagi-Uda antenna with eleven elements is shown in Fig. 8.

MATLAB program

MATLAB R2008b or higher versions can be used for the program. The step-by-step procedure for program testing follows:

1. Run MATLAB from Desktop
2. Open 'Expt_A1_2D.m' file from 'File' menu
3. Import 'Polalist-yagi uda.txt' file

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To obtain the radiation pattern of a Yagi-Uda antenna in MATLAB pro-

gram, a table with theta (or θ) values for polar angles and corresponding amplitudes of radiation patterns is required. These values can be obtained using antenna trainer. The data table should be saved in text (.txt) format.

from 'File' menu by selecting "Import Data" option

4. Click 'Next' button
5. Click 'Finish' button
6. Compile and run the program from 'Debug' menu by clicking 'Run' option
7. Go to the main MATLAB window. The command window will prompt you to enter the offset value for theta
8. Enter value like 90, 80 or anything as per your requirement and then press 'Enter' key
9. You should get the directivity output plots in polar as well as in cartesian forms

On entering the appropriate value, the program gives a good directivity plot. Maximum directivity is 4.483 or 6.5157 dB.

EFY note. The source codes and data table of this project are available on www.efymag.com. ●

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